Physicist / Engineer physicist BSc, year III / semester 1
"Optics problem solving"

MIDTERM TEST 1

30-10-2025

1 Geometrical optical analysis of a metalens with the help of Fermat's principle

You are examining a metasurface in geometrical optical approximation at $\lambda_0 = 1550\ \mathrm{nm}$ wavelength, see Fig. 1. This alters the phase of the traversing light with the help of nanostructures patterned in different ways, which is taken into account as if the surface were continuous, but the refractive index would change in it as a function of position. Let the effective refractive index distribution be cylindrically symmetric: $n(r) = 2.1 - 17.6 \cdot r^2$, with $r \in [0, 0.25\ \mathrm{mm}]$, and the thickness of the amorphous silicon layer comprising the metasurface be $d = 1\ \mu\mathrm{m}$. A plane wave propagating parallel to the optical axis is incident at the metalens from the left. In terms of Fermat's principle determine that r height, at which the passing light ray intersects the optical axis at a distance of $z = 28.4085\ \mathrm{mm}$ in air! The metasurface is so thin, that we neglect light refraction inside of it. The refractive index of the silica be n_s. Use parametric calculations as long as it is possible! **25 p**

$$OPL(r) = n_\mathrm{s} z_\mathrm{w} + n(r)d + \sqrt{r^2 + z^2} \tag{1}$$

$$OPL(r) = n_\mathrm{s} z_\mathrm{w} + (2.1 - 17.6 \cdot r^2)d + \sqrt{r^2 + z^2} \tag{2}$$

$$\frac{\mathrm{d}OPL(r)}{\mathrm{d}r} = -17.6 \cdot 2rd + \frac{1}{2}\frac{1}{\sqrt{r^2+z^2}}2r = 0 \tag{3}$$

$r = 0$-trivial solution.

$$35.2d = \frac{1}{\sqrt{r^2 + z^2}} \tag{4}$$

$$r^2 = \frac{1}{(35.2d)^2} - z^2 \tag{5}$$

$$r = 0.1832\ \mathrm{mm}. \tag{6}$$

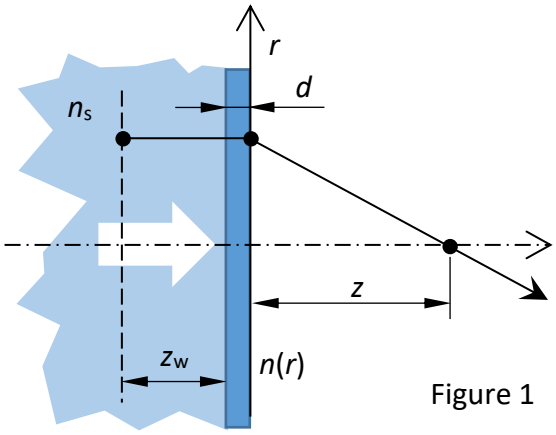


Figure 1

2 Modelling the beam propagation of a laser guide star as a Gaussian beam

The ESO observatory on Paranal Mountain (VLT) uses a laser guide star to determine wave-front distortions caused by atmospheric turbulence. For this reason, sodium ions being in the mesosphere at a distance of $z = 95\ \mathrm{km}$ from the surface are excited by a laser tuned to $\lambda_0 = 589\ \mathrm{nm}$. Propagation of laser light is approximated by a Gaussian beam, whose waist is at the emitter telescope, its radius here is $w_0 = 180\ \mathrm{mm}$. What is the Rayleigh range (z_R)? What is the beam radius in the mesosphere $w(z)$? If we denote peak intensity here by I_0, how can we

express intensity as a function of distance (r) measured form the optical axis? Based on this what is the full-width-at-half-maximum (D_{FWHM}) of the light spot excited in the mesosphere? What is the angle $(\Delta\theta)$ in arc seconds subtending this as viewed from the surface? **25 p**

$$z_{\mathrm{R}} = \frac{\pi w_0^2}{\lambda_0} = \frac{\pi \cdot 0.180^2}{589 \cdot 10^{-9}} = 172.8\ \mathrm{km} \tag{7}$$

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{z_R}\right)^2} = 205.4\ \mathrm{mm} \tag{8}$$

$$I(r) = I_0 e^{-2\left(\frac{r}{w}\right)^2} \rightarrow e^{-2\left(\frac{D_{\mathrm{FWHM}}/2}{w}\right)^2} = 0.5 \tag{9}$$

$$D_{\mathrm{FWHM}} = 2w \cdot \sqrt{\frac{\ln 0.5}{-2}} = 241.8\ \mathrm{mm} \tag{10}$$

$$\Delta\theta = 2\,\mathrm{atan}\frac{D_{\mathrm{FWHM}}}{2z} = 0.525'' \tag{11}$$

3 Describing reflection of a lithographic chromium mask with complex refr. index

You are examining a chromium mask used in photolithography in case of a plane wave at normal incidence with a wavelength of $\lambda_0 = 632.8\ \mathrm{nm}$, the outside medium is air. A chromium layer of $d = 100\ \mathrm{nm}$ thickness is deposited on a dielectric substrate (glass) according to Fig. 2. The real and imaginary parts of the complex refractive index of chromium are: $n' = 3.140$ and $\kappa' = 3.315$. What are the amplitude and phase of the complex reflection coefficient $(\tilde{\rho})$ at the outside surface of chromium? How much is the reflectance (R)? How much is the penetration depth (δ)? What is the phase difference $(\Delta\varphi)$ between the wave reflected off the chromium surface and the glass surface next to the chromium layer, considering phase shifts caused by reflection on chromium and glass too? Measure the phase from the wavefront arriving at the chromium surface. **25 p**

$$\tilde{\rho} = \frac{n - \tilde{n}'}{n + \tilde{n}'} = \frac{1 - n' + i\kappa'}{1 + n' - i\kappa'} = \frac{-2.14 + i3.315}{4.14 - i3.315} = \frac{3.946 \cdot e^{i2.144}}{5.304 \cdot e^{-i0.6752}} \tag{12}$$

$$\tilde{\rho} = 0.7440 \cdot e^{i2.819} = -0.7056 + i0.2357\ ;\ \ 161.5° \tag{13}$$

$$R = |\tilde{\rho}|^2 = 55.35\% \tag{14}$$

$$\delta = \frac{1}{k_{\mathrm{im}}} = \frac{\lambda_0}{2\pi\kappa'} = 30.38\ \mathrm{nm} \tag{15}$$

$$\Delta\varphi = 2.819 - \left(OPL\frac{2\pi}{\lambda_0} + \pi\right) = 2.819 - 2d\frac{2\pi}{\lambda_0} - \pi \tag{16}$$

$$\Delta\varphi = 2.819 - 1.986 - 3.142 = -2.309\ \mathrm{rad} = -132.3°. \tag{17}$$

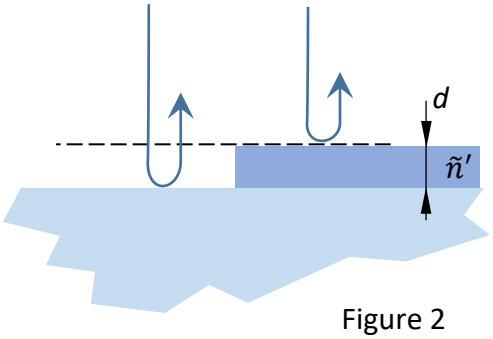


Figure 2

4 Analysis of a variable focal length lens system with matrix optics

Off-the shelf optical lenses have a typical effective focal length uncertainty of $\pm 1\%$, but for imaging tasks it is often required to adjust the effective focal length of a lens system with higher precision. To achieve this, the simplest way is to implement the system with two lenses, and fine-tune the resulting effective focal length by the distance between the lenses. Let us have two lenses designed to be identical with effective focal lengths of $f_1 = 90.9\ \text{mm}$ and $f_2 = 89.1\ \text{mm}$, and model these as thin lenses. What are the refractive powers of the lenses (P_1 and P_2)? Write up the transfer matrix $\mathbf{M}$ of the system from the entrance surface of the first lens to the exit surface of the second with parameters (the lenses are located in air)! How can we get from this the expression of the resultant focal length? How should we select distance d between the lenses to have $f = 50.0\ \text{mm}$ focal length? What are the $ABCD$ coefficient values of matrix $\mathbf{M}$? Where is the image-side principal plane of the system relative to the plane of the second lens (z')? **25 p**

$$P_1 = \frac{1}{90.9} = 0.01100\ \text{mm}^{-1} \quad ; \quad P_1 = \frac{1}{89.1} = 0.01122\ \text{mm}^{-1} \tag{18}$$

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ -P_2 & 1 \end{bmatrix} \begin{bmatrix} 1 & d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -P_1 & 1 \end{bmatrix} \tag{19}$$

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ -P_2 & 1 \end{bmatrix} \begin{bmatrix} 1 - dP_1 & d \\ -P_1 & 1 \end{bmatrix} \tag{20}$$

$$\mathbf{M} = \begin{bmatrix} 1 - dP_1 & d \\ -P_1 - P_2 + dP_1P_2 & 1 - dP_2 \end{bmatrix} \tag{21}$$

$$f = \frac{1}{P_1 + P_2 - dP_1P_2} \tag{22}$$

$$fP_1 + fP_2 - fdP_1P_2 = 1 \tag{23}$$

$$d = \frac{fP_1 + fP_2 - 1}{fP_1P_2} = 18.02\ \text{mm} \tag{24}$$

$$\mathbf{M} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 0.8018 & 18.02 \\ -0.02 & 0.7978 \end{bmatrix} \tag{25}$$

$$z' = (1 - A)\frac{1}{C} = -9.91\ \text{mm}. \tag{26}$$